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Subsea Repair and Decommissioning Optimized Solutions

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Abstract

Objectives/Scope: This presentation includes a review of the principles and solutions for efficient repair of pipelines, flowlines and risers related to cost-effectiveness when considering ESG and sustainability. The various possibilities considering repair versus decommissioning are handled with focus on optimized solutions.

Methods, Procedures, Process: IK-Subsea has for more than 37 years of operation developed several tools, solutions and products for the operators with subsea assets. The design and solutions have mostly been related to repair of old flowlines, pipelines and riser, and also equipment for recovery and decommissioning. During the recent years, these activities have increased in extent and numbers with several new challenges related to efficient and safe operations and issues related to spillage. Therefore, IK-Subsea have developed several new tools focusing on avoiding spillage and enabling quicker and more efficient repair or recovery of pipelines, flowlines and risers.

Results, Observations, Conclusions: The need for safe recovery of old facilities filled with polluted elements / hydrocarbons can be performed safely by use of specialized isolation tools, such as plugs and caps. These tools are typically operated by ROVs with the adherent operational requirements. The required recovery operation such as lifting of assets to be decommissioned have several challenges as the integrity of the assets itself will be questioned. This typically apply for old flowlines, flexible and risers where the tool design need to consider this weak element of the recovery operations. The repair solutions used in the late life of the assets need to consider the decommissioning to be performed. However, the best and most optimal approach of handling the old lines is to continue the operation by continuously monitoring the integrity and perform the preventive measures as required. This is the optimal approach considering both the environmental and financial aspect. (ESG and sustainability)

Novel/Additive Information: During the late life operation of subsea assets, the evaluations related to extending the lifetime by performing the required repair operations is considered versus the ongoing operational cost and then the later decommissioning to be performed. This presentation will give the audience additional understanding of the possibilities that these solutions can optimize the late life field operations.

Introduction

The infrastructure and facilities that is designed and manufactured for the production of oil and gas industry is typically designed with a lifetime of 25 years. Sometimes much longer, but very often shorter than the actual lifetime that is required when the field-lives are extended.

Therefore, there are several pipelines, flowlines and risers that are operated beyond the original design lifetime.

For some cases the facilities that are beyond the lifetime, new facilities are installed, and the old facilities are decommissioned. For other cases, the facilities are still in operation, however with an inspection and maintenance program that will secure sufficient safety and integrity for further operation.

This paper is elaborating the late life / beyond lifetime operation issues for subsea pipelines.

IK-Subsea is a niche company for subsea pipeline repair, and that is the basis for this paper.

This paper's focus is "Optimized Solutions" Traditionally this would mean the most cost-efficient solution. In this paper we will consider an EGS approach related to the word "optimized"

The definition of ESG can be:

Environmental, social, and governance for an investing principle that prioritizes environmental issues, social issues, and corporate governance. Investing with ESG considerations is sometimes referred to as responsible investing or, in more proactive cases, impact investing.

An alternative for an ESG approach is to consider "Sustainability" The definition of this can be:

Sustainability is a social goal for people to co-exist on Earth over a long period of time. Definitions of this term are disputed and have varied with literature, context, and time. Sustainability usually has three dimensions: environmental, economic, and social.

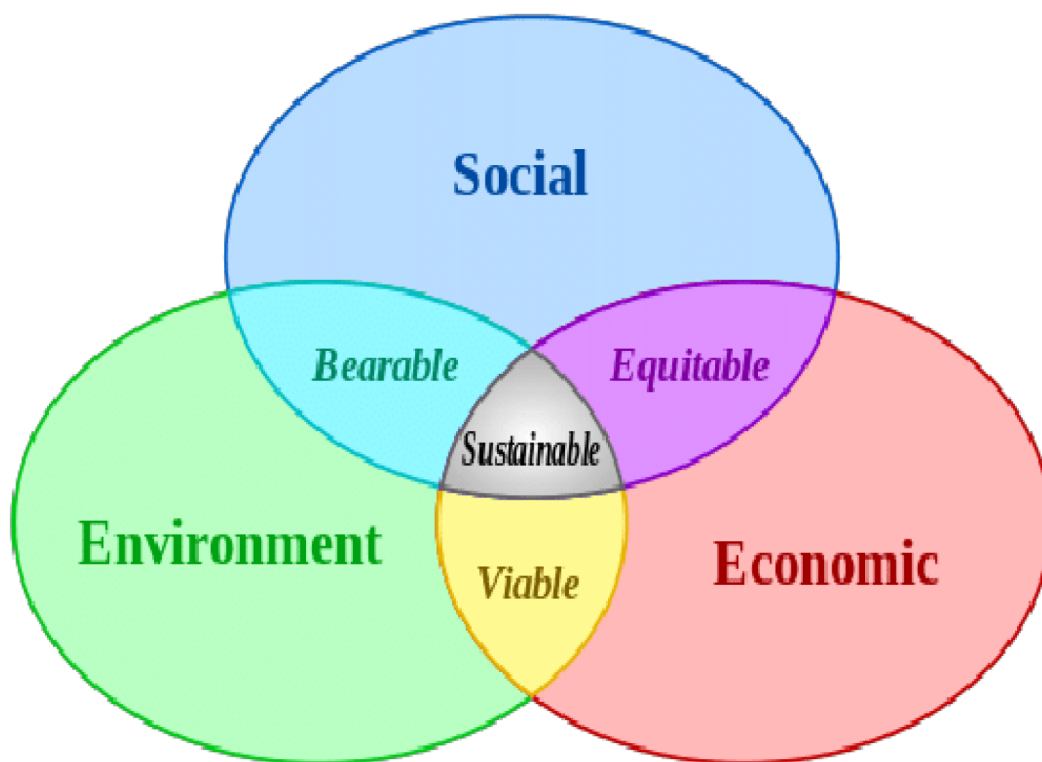


Figure 1—Illustration of environmental, economic and social

So, what does this approach mean for IK as a small supplier in the oil and gas business. IK is niche supplier, but we have our contribution and our goals in the pipeline operations, and the below set-up illustrates how we see our self in the ESG and Sustainability relation:

IK-Subsea's core business in short is very simplified the following:

- Prevent Leakages
- Stop leakages

Our services shall then contribute with the following three ESG goals for the very best or optimized solutions:

IK-Subsea – Sustainability – Social/Safety Focus

- Safe operations
- Sufficient Integrity of the System and Solutions

IK-Subsea – Sustainability – Operator/Cost Focus

- Delivery cost-efficient repair solutions
- Keep the lines running safely in a cost-efficient way
- Extend the pipelines lifetimes beyond the original lifetime
- Assist during the decommission of pipelines

IK-Subsea – Sustainability – Environmental Focus

- Tail Production Partner – Secure that all resources are recovered
- Save CO₂ by preventing manufacturing of new lines
- Design of CO₂ efficient Clamps
- Safe decommissioning with no spillage
- The cost savings to be used for developing green energy

The three above ESG parts are the three guidelines for the content of this paper. In order to make it align with our services and solutions we have organized the separate sections as follows.

The pipeline Repair Market, Typical Repair Solutions, The Integrity of the Solutions and Systems, The Environmental Impact,

The Economical Impact and Conclusion

The Pipeline Repair Market

As described in the Introduction there are two main factors that drives the pipeline repair market:

- All facilities get one year older every year
- Several pipelines are operated far beyond the original lifetime

The above is simple and logic and can be easily illustrated with the below graph showing the number of subsea pipeline clamps delivered by IK of the last years.

The clamps have been delivered for both corrective applications and preventive applications. Both cases are contributing to sustainability. For the preventive applications the clamp solution is definitely in line with the "sustainability" definition. For the corrective applications, which normally means stopping a leakage,

the clamp stopping the leakage can be regarded as sustainable, but the operator has failed by not having a sustainable operation of their facilities.

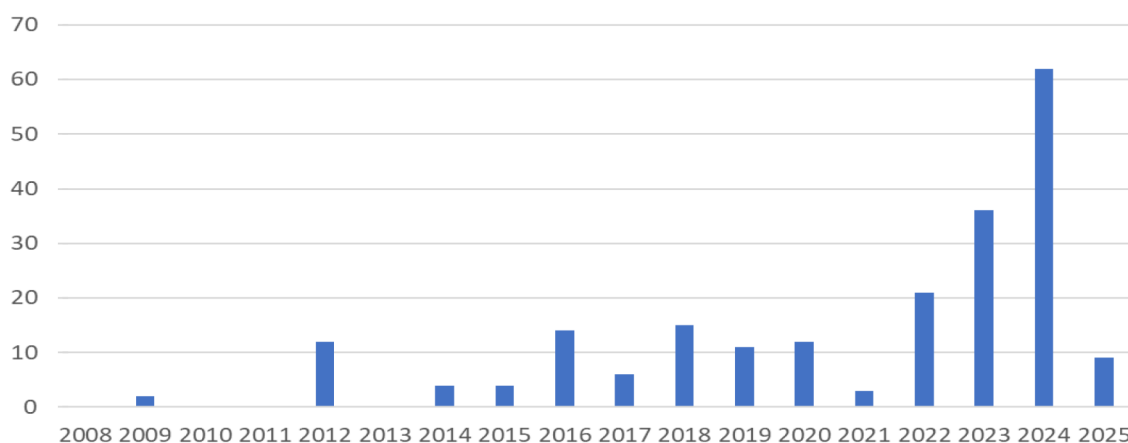
The total of appr 200 clamps delivery by IK over last years the number of corrective and preventive deliveries are as follows:

- Corrective cases / clamps 28 locations
- Preventive cases / clamps 170 locations

The 28 corrective cases, all related to leakages are 28 too much! The cost for the operators, society and environment are far higher for these incidents, than for the corrective deliveries. In general, a prudent operator shall not face the need for corrective repair operations.

This paper does not look further into the costs related to corrective repairs other than stating that it is normally extremely expensive, not only financial but also related to company reputation and environmental impacts which can be devastating.

Table 1—Number of repair clamps delivered from 2009 to 2024



Typical Repair Solutions

As described in previous section there are both corrective repair solutions and preventive solutions. In general, the solutions will normally be the same, as the preventive solutions are designed for leakages that is at risk to occur.

However, the related cost and installation time is far more for the corrective solutions, and a prudent operator shall of course avoid the need for corrective repairs, this is closer described in the next section.

This section is included for the reader to understand the pipeline repair clamp concept. The general concept is that a defected section of a pipeline is covered up with an external two parted / split spool that is sealing towards the pipeline. By that the clamp replaces the pipeline function at the defected location. This is more or less the universal description of the pipeline repair clamp.

Clamps have been used for repair of pipelines for several decades, both subsea and topside / onshore.

There are several suppliers and there are several different designs. However, there are a couple of design principles that are general / typical for such repair clamps. Then there are two main groups that again can be split into several sub-sections (but we shall not enter that path)

The two main groups are:

- The local defects clamp family, used for local pipeline defects such as cracks, pits and weld issues.
- The split clamp family, used for larger sections of defects such as general corrosion or long longitudinal cracks

These two groups are again split into two sections.

- The sealing clamps, used for pipelines with defects with the structural integrity intact.
- The structural sealing clamps, used for pipelines with defects and the structural integrity is jeopardized.

A split sealing repair clamp consists of two clamp halves connected by a set of main bolts. Compression packers interacting with face seals provides sealing for the clamp cavity. Actuator rings provide pressure to the compression packers. Face seals are passive (O-ring type).

The local defect clamp family is based on the same principle, however, as the sealing is for a local defect there is one compression packer only, that seal directly onto the defect location.

Both types of clamps use toothed slips technology for obtaining structural reinstatement of the pipeline, if the defects are severe.

The two main types with the two sub types of clamps are illustrated on the next pages.

The local defect clamps family is illustrated on the below two sketches. This clamp is very installation friendly with all the required setting tools pre-mounted, making it very ROV friendly.

The below sketch shows the clamp being set onto the defected pipeline. The black element inside the clamp is the compression packer that is going to be closed around the pipe when the clamp is closed and the packer compressed by the setting ring activated by the setting bolts and torque tools.

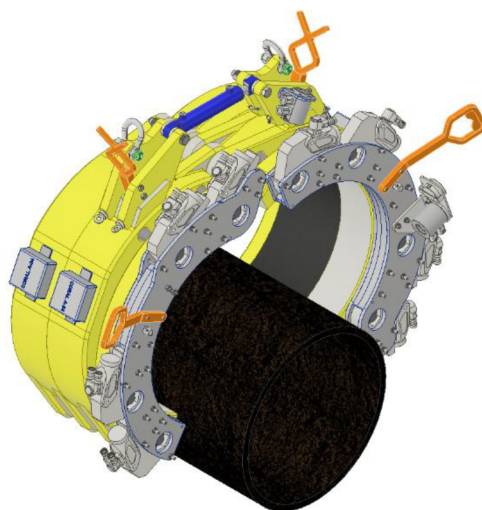


Figure 2—The local defect clamp with tooling package

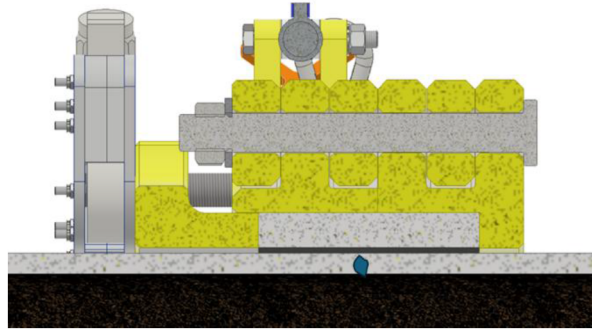


Figure 3—The local defect clamp Cross Section with the setting ring torque tool

The local defect integrity clamp is based on the same ROV friendly installation principle and it is also designed for local defects, however with teathed slips to reinstate the structural capacity at the defect location, that is obtained by calibrated torque tools and controlled lubrication of the setting bolts threads.



Figure 4—The Compact Integrity Clamp with tooling package.

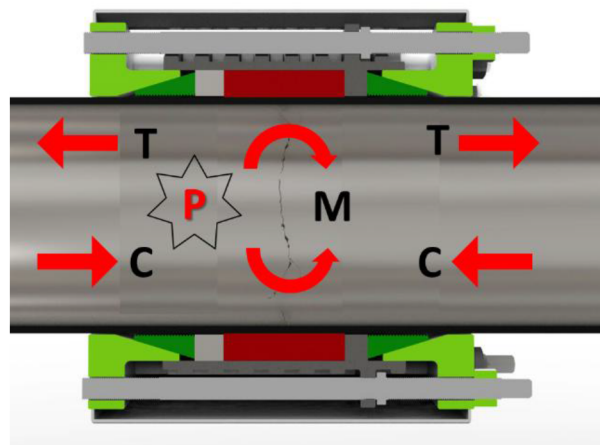


Figure 5—The Compact Integrity Clamp Cross Section with the external loads illustrated with the arrows. The clamp can handle any external loads that are imposed on the defect location

Figures below shows the main components, assembly, and key dimensions of a typical split repair clamp.

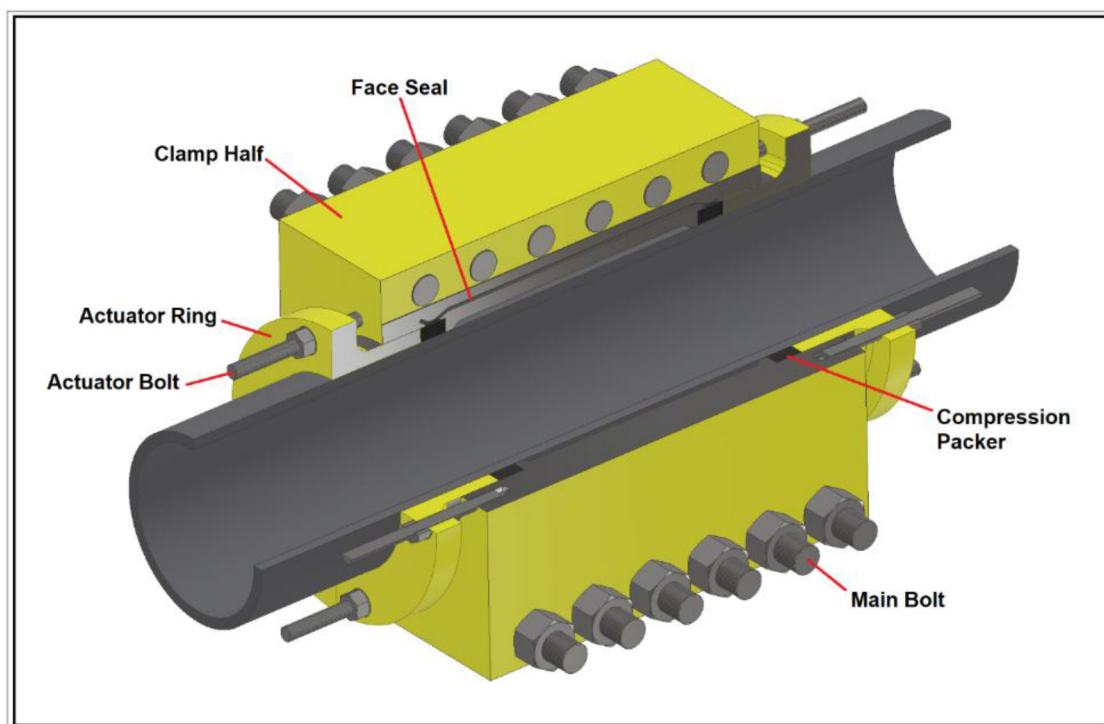


Figure 6—Split Repair Clamp Components

The below sketch illustrates the Split Sealing Clamp. The length / Size of the cavity in the middle is dimensioned to envelope the defect location. The length may vary from the half pipeline diameter to a length of several pipeline diameters. Clamps with length up to 4 – 5 meters

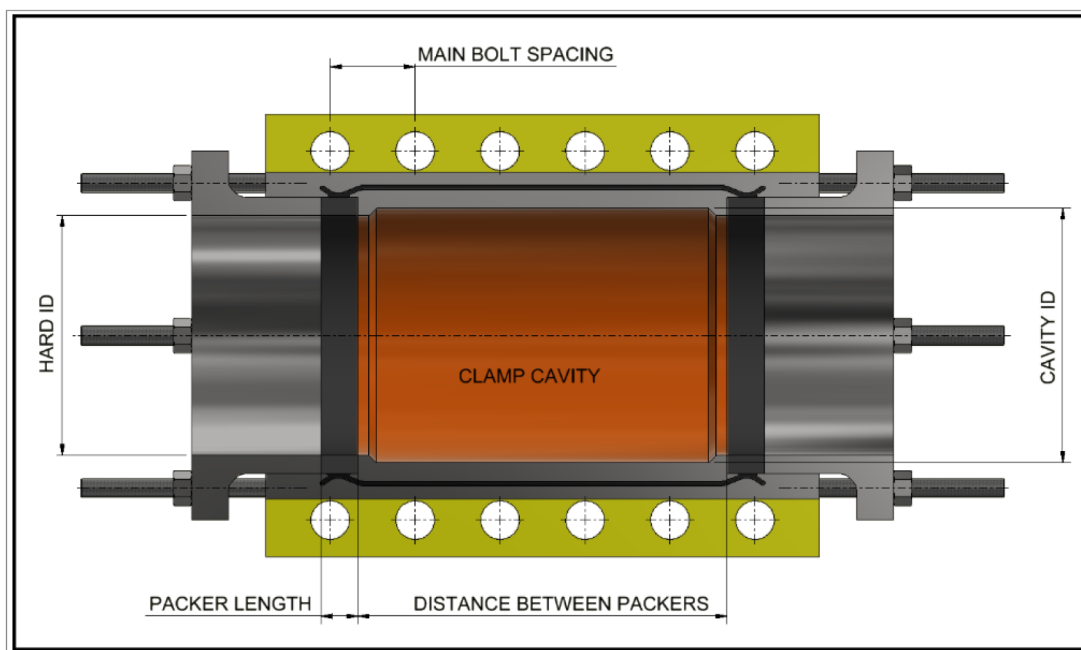


Figure 7—Clamp Key Dimensions

The structural split clamps has the same basis layout as the sealing split clamp, however with teathed slips in order to mechanically fix / connect the pipeline and the clamp and then transfer the external loads in the pipeline through the clamp at the defect location.

The below sketch illustrates structural sealing clamps with the main sealing elements and the slips, the two key elements of a structural sealing clamp.

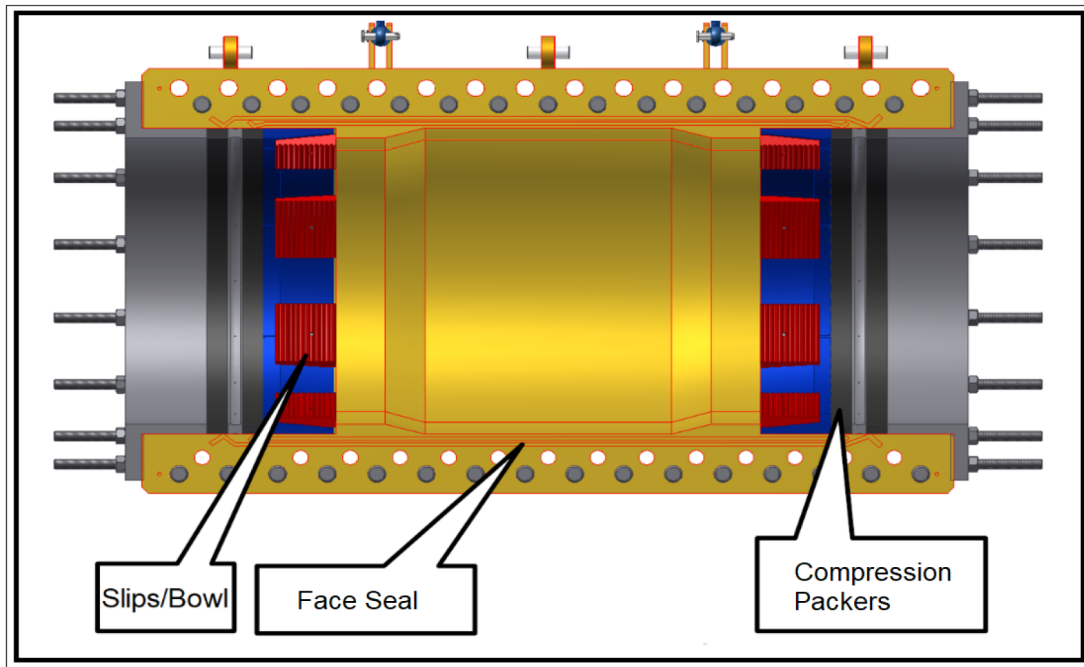


Figure 8—Structural Seal Clamp

The Integrity of the Solutions and Systems

As described previously in this tender, and optimized operation requires that the equipment / pipelines have the sufficient integrity for the operational parameters.

This integrity needs to be handled, monitored and controlled by sufficient means.

As an example, a "brand new pipeline" with perfect inspection results shall have an inspection program that is adapted to that. The inspection frequency can be low.

On the other hand, an "old pipeline" with issues and defects needs to be regularly inspected to confirm that the findings in the pipeline are within acceptable limits. The pipeline is old and getting close to the final decommissioning, but there are still years of operation to come. For such pipelines the "Circle of Integrity" is an efficient and required approach for securing the operation.

The definition of the Circle of Integrity is:

Activities which ensure that equipment and systems reliably perform in accordance with their original or extended design life with minimal outage and at minimum cost, while causing no harm to health, environment or property.

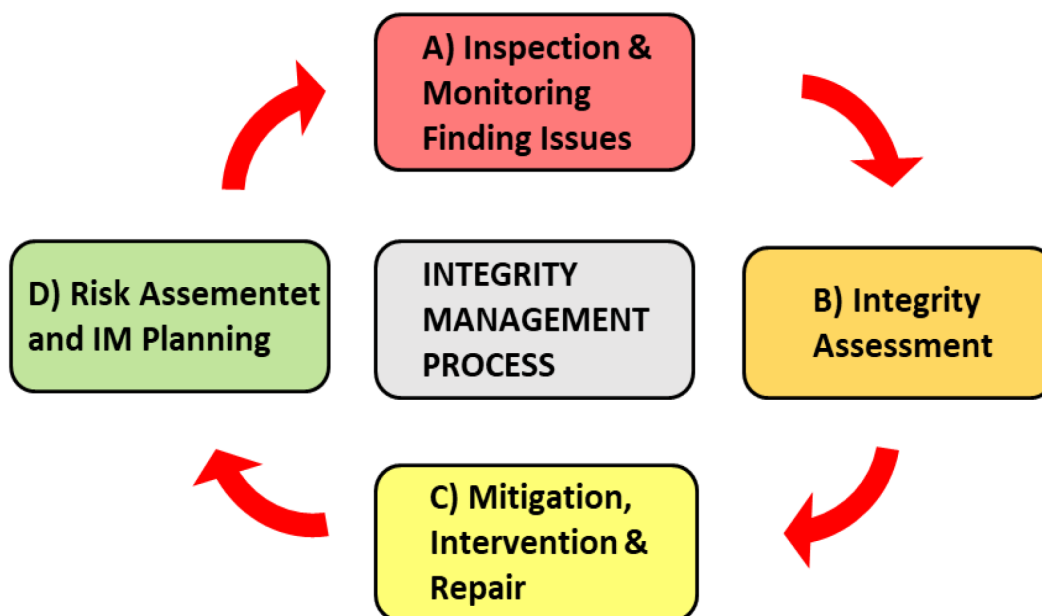


Figure 9—The Circle of Integrity

The above figure illustrates the Circle of Integrity process. It is an ongoing process, but the "rotational" speed will vary. The 4 activities are closer described in the below sections.

- A. Findings which may require repair is revealed during inspection.
- B. The findings are evaluated by means of engineering assessments and evaluated against short and long terms risks and potential for further negative development.
- C. If found required, emergency and permanent repair solutions are planned and installed. The repair solutions to be documented in accordance with governing design rules and regulations including documentation for operation.
- D. The functionality and integrity of the repaired solution to fulfill functional requirements for the remaining lifetime by defining requirements for inspection and maintenance.

The Circle of Integrity requires involvement of several parts / stakeholders, from the owner of the pipeline to the operator and then the various service companies that have different tasks and deliveries required for keeping the pipeline safe and sound (sufficient integrity) IK, as the repair company will have the following tasks and focus in this circle. The findings which will require a repair will be carried out in a repair project with a robust project management where all relevant activities are included. The main focus for a typical repair project is to reduce the risk for escalating failure by means of installing a rapid emergency repair solution and to ensure that this repair solution is being further developed into a long-term repair solution. The repair solution will be fully documented for design and operation which ensure the required function and safety level for the repaired facility for the remaining lifetime:

- Damage evaluations including review of existing design and integrity documentation, inspection and survey reports and execution of preliminary analyses.
- Evaluate emergency and permanent repair solution
- Carry out preliminary risk assessments
- Design development of the selected repair solutions including local and global analyses
- Design, manufacturing, fabrication, assembly, testing and installation

- Documentation for design and operations in accordance with relevant design requirements
- Defining requirements for risk-based inspection and required maintenance for the entire service life

The Environmental Impact

There are several stakeholders and regulatory authorities that currently are launching new rules and regulation related to environmental issues and impacts.

Then there are practical and operational approaches that can benefit the environment and at the same time benefit the finances. The environmental issues are discussed in this section and in the next section a concrete cost example comparing pipeline repair versus new built is included.

The Corporate Sustainability Reporting Directive (CSRD)

This is a report published by companies on the environmental, social and governance (ESG) impacts of their activities. It enables addressees and users to understand more clearly the impacts of a company's business activities on the environment and society and to assess the risks and opportunities companies face, or which are offered to them. It is a communication tool that plays an important role in convincing skeptical observers that the company's actions are sincere.

The growing importance of sustainability reports is due to the fact that investors and other stakeholders are calling on companies to disclose more information about their sustainability activities and environmental, social, and governance strategies.

Many new legislative documents on ESG requiring companies to disclose sustainability information have already become effective or are currently being prepared:

- REGULATION (EU) 2020/852 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 18 June 2020 on the establishment of a framework to facilitate sustainable investment (EU Taxonomy Regulation, in force since July 2020)
- Corporate Sustainability Reporting Directive (CSRD, in force since January 2023)
- Directive on Corporate Sustainability Due Diligence (CSDD, draft)
- Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL establishing a framework for setting ecodesign requirements for sustainable products

IK initiative for reducing Environmental Impact

IK have together with a climate advisor consultant company reviewed all the process and activities in the company in order to both map and to reduce the total CO₂ emissions from the complete relevant value chain.

The results have been an appr 30% reduction of the CO₂ emission from the ongoing business, this is in the range of 3000 kg CO₂ emission. Considering that IK in these days in average deliver appr 50 clamps per year, this reduction is in the range of 150 000 kg CO₂. This is not a large volume when taking a global approach. But it is a good example for other companies and can be significant when all similar suppliers as IK do the same process.

The process has included a total mapping of all IK activities and also the value chain. This includes design and manufacturing as well as the everyday domestic related tasks as waste handling, light and temperature....etc....etc.

This activity has also impacted positively of the financial results.

Benefit from postponing the Decommissioning

The optimized solution for a pipeline is to keep it up and running with sufficient integrity for the entire period required. With other words, to postpone the decommissioning as far out as possible.

The decommissioning operation itself is not environmentally friendly, requiring large vessels and operations with long durations. At the same time the marine environment will be significantly disturbed at the seabed pipeline route.

All pipelines will at the very end be decommissioned, however, the best way related to the environment has been a discussion for many years. There are different rules and practices in the different regions. In some regions pipelines over a defined diameters are allowed to be left at the seabed, other locations have other rules.

It is likely to believe that the optimized decommissioning philosophy is still to be developed as the pipeline operators gain more knowledge. The best way with less environmental impact is still to be settled.

Based on the above, postponing the decommissioning is environmental beneficial.

Extended operational life of the pipelines

Pipelines are designed for a defined lifetime. When or if the pipeline is intended for extending the lifetime due to the production passing to the original production plan profile, new or extended operational authorizations need to be received from the Authorities.

This is a standard worldwide practice including several thousand of kilometers of subsea pipelines. The integrity becomes a more critical issue as the age increases, but as described in the previous Integrity Section, applying adequate inspection and control programs will secure safe and sound operation.

Then there need to be repair methods with proven integrity available on the market. There are several worldwide subsea IMR companies (Inspection Maintenance Repair) and then are several suppliers of pipeline repair equipment, installation equipment and remedial tools as pipeline clamps.

The alternative for repair of a pipeline passed the design lifetime is installation of a new. The environmental impact from new built or repair is exemplified on the next page.

These numbers and the input are simplified and are used for giving the reader an overview and not precise numbers. There are numerous factors and parameters required in order to give correct number. The numbers used are based on typical input for typical cases and the message is clear.

The following sources have been used for the estimates of the CO₂ emissions on the next page:

- Steel production: Terravera Foundation report
- Clamp production: IK-Norway AS / Climate Advisor AS report
- Offshore fuel consume: North Sea Marine Contractor

CO₂ Emission example, New pipeline versus repair

Defects to be corrected on a 20 km subsea 16" 100 bar Oil Pipeline ROV operation.

The below numbers indicate the CO₂ impact from replacing the old 16" pipeline with a new or repair with several alternative clamp solutions and number of clamps.

As can be seen in the below tables, even the most serious repair scenarios will result in a CO₂ emission that is just a few percent of the pipeline replacement scenario.

Installation of new line;	5 100 Te CO₂
Manufacturing of Pipeline, steel: 150 Te per km × 1,5 Te CO ₂ per Te × 20 km	4 500 Te CO ₂
Pipelaying, fuel consumption: 15 Te per day × 7 days × 2,7 Te CO ₂ per Te	300 Te CO ₂
Misc, transport, steaming, coating layers, other services...etc.	300 Te CO ₂

CO2 Emission Examples, 4 Types of Repair Scenarios, 1 or 10 installations

Table 2—CO2 Emission for Clamp Repair – 1 clamp installation

Defined defects and repair methods, 16" pipeline, 100 bar & ROV operation		Estimated CO2 Emission - 1 defect location		
Type of defect	Type of repair clamp	Design, Manufacturing & Testing	Vessel Mob/Demob & Operation	Total CO2
Local defect, sealing	Compact Sealing Clamp	2,5	20	20
Local defect, sealing + structural	Compact Integrity Clamp	3,5	20	20
Extensive defect 2 m, sealing	Split Sealing Clamp	4,0	25	25
Extensive defect 2 m sealing + structural	Split Structural Sealing Clamp	6,0	30	30

Table 3—CO2 Emission for Clamp Repair – 10 clamp installations

Defined defects and repair methods, 16" pipeline, 100 bar & ROV operation		Estimated CO2 Emission - 10 defect locations		
Type of defect	Type of repair clamp	Design, Manufacturing & Testing	Vessel Mob/Demob & Operation	Total CO2
Local defect, sealing	Compact Sealing Clamp	20	80	100
Local defect, sealing + structural	Compact Integrity Clamp	30	80	110
Extensive defect 2 m, sealing	Split Sealing Clamp	35	100	135
Extensive defect 2 m, sealing + structural	Split Structural Sealing Clamp	50	150	200

The Financial Impact

IK have had a complete review of our processes in order to reduce the CO2 emission and the cost. What we have learned that is the reduced CO2 emission and reduced cost correlates very well for most of the measures that have been reviewed and revised.

The focus has been to reduce the consumption of resources. The resources mean; material, energy and manhours.

Focus areas during design for improved sustainability:

- Use as little material as possible / low weight.
- Reduce the amount of welding
- Reduce the amount of machining

- Do not use tighter tolerances that is required
- Reduce the waste of material
- Do not define higher material quality than required
- Focus on simple and easy assembly
- Consider the test method and the test equipment
- Consider the installation operation, reduced time
- Reduce the need for transport

The results have been an appr 25% reduction of the complete cost compared with the ongoing business cost level. This is significant and give the double positive effect as the environment solution can compete in the market and by the use contribute to the general CO2 emissions



Photo 1—The young IK team working with the improvements of the clamp production.

The following sources have been used for the estimates of the cost elements on the next page:

- | | |
|------------------------|-----------------------------|
| • Pipeline production: | North Sea Operator |
| • Clamp production: | IK-Norway AS |
| • Offshore cost: | North Sea Marine Contractor |

Repair versus new build – Example with cost estimates

Defects to be corrected on a 20 km subsea 16" 100 bar Oil Pipeline ROV operation. The same pipeline as in previous section considering the CO₂ emissions.

The below numbers indicate the cost impact from replacing the old 16" pipeline with a new or repair with several alternative clamp solutions and number of clamps.

As can be seen in the below tables, even the most serious repair scenarios will result in a very positive financial result.

Installation of new line cost,**35 mil USD**

Purchasing of Pipeline:	1 mill USD per km	20 mill USD
Pipelaying:	0,5 mill USD per km	10 mill USD
Vessel Mob / Demob:		5 mill USD

Table 4—Estimated cost for Clamp Repair – 1 clamp installation

Defined defects and repair methods, 16" pipeline, 100 bar & ROV operation		Estimated cost in USD - 1 defect location		
Type of defect	Type of repair clamp	Design, Manufacturing & Testing	Vessel Mob/Demob & Operation	Total Cost
Local defect, sealing	Compact Sealing Clamp	250,000	400,000	650,000
Local defect, sealing + structural	Compact Integrity Clamp	400,000	400,000	800,000
Extensive defect 2 m, sealing	Split Sealing Clamp	500,000	500,000	1,000,000
Extensive defect 2 m, sealing + structural	Split Structural Sealing Clamp	700,000	600,000	1,300,000

Table 5—Estimated cost for Clamp Repair – 10 clamp installations

Defined defects and repair methods, 16" pipeline, 100 bar & ROV operation		Estimated cost in USD - 10 defect location		
Type of defect	Type of repair clamp	Design, Manufacturing & Testing	Vessel Mob/Demob & Operation	Total Cost
Local defect, sealing	Compact Sealing Clamp	1,500,000	1,000,000	2,500,000
Local defect, sealing + structural	Compact Integrity Clamp	2,500,000	1,000,000	3,500,000
Extensive defect 2 m, sealing	Split Sealing Clamp	3,500,000	1,400,000	4,900,000
Extensive defect 2 m, sealing + structural	Split Structural Sealing Clamp	5,500,000	1,600,000	7,100,000

Conclusion

In this paper we have tried to enlighten and discuss the benefits and issues related to pipelines to be operated past the original lifetime.

The issue has been discussed with focus on ESG and Sustainability, then with the concrete focus on CO2 emissions and finance, but also considering the safety by elaboration around the "Circle of Integrity"

The content should be regarded as input for discussion and for decisions to be taken by operators and contractors when making plans for pipeline operations extending the original life time.

The general conclusion of this paper is that extended lifetimes and operations of pipelines is beneficial considering both the financial and the environmental part. This is however based on simplified approaches, there are a numerous of factors that has not been included in this paper that need to be considered for making the correct decisions when extending the life of old pipelines.

The main message that we believe is significant and relevant is that cost reduction and reduced CO₂ emission can be directly correlated for both the suppliers and the operators.